THE RELATIVE CUP PRODUCT IS COMPATIBLE WITH COMPOSITION OF MORPHISMS OF RINGED TOPOI

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the adjoint description of both composites around the displayed diagram and the identification of the composed counit through the categories lemma.

Editorial note. The label lemma-compose-cup-product names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement's displayed commutative diagram is the author's and is delivered verbatim; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The relative cup product and its adjoint description. Same chapter, the remark constructing the cup product Rf_*K tensor^L Rf_*L -> Rf_*(K tensor^L L) as the adjoint of the canonical map on pullbacks. Supplies the maps appearing in the displayed diagram.
- Composition of counits. Categories, the lemma stating that the composition of the counits for f and f' is the counit for the composite. Supplies the identification of the two composites as the same adjoint map.

Source reference. Original lemma 0DV0: sources/stacks-chow/tags.txt, source lines 17345–17345. Prerequisite pointer invoked at proof line(s) 8525; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0B68: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4193–4193. Prerequisite pointer invoked at proof line(s) 8488; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : (Sh(\mathcal{C}), \mathcal{O}) \to (Sh(\mathcal{C}'), \mathcal{O}')$ and $f' : (Sh(\mathcal{C}'), \mathcal{O}') \to (Sh(\mathcal{C}''), \mathcal{O}'')$ be morphisms of ringed topoi. The relative cup product of Remark 0B68 is compatible with compositions in the sense that the diagram*

$$\begin{array}{ccccc} R(f' \circ f)_* K \otimes^{\mathbf{L}}_{\mathcal{O}''} R(f' \circ f)_* L & = & & = & Rf'_* Rf_* K \otimes^{\mathbf{L}}_{\mathcal{O}''} Rf'_* Rf_* L \\ \downarrow & & & & \downarrow \\ R(f' \circ f)_*(K \otimes^{\mathbf{L}}_{\mathcal{O}} L) & = & Rf'_* Rf_*(K \otimes^{\mathbf{L}}_{\mathcal{O}} L) & \leftarrow & Rf'_*(Rf_* K \otimes^{\mathbf{L}}_{\mathcal{O}'} Rf_* L) \end{array}$$

is commutative in $D(\mathcal{O}'')$ for all K, L in $D(\mathcal{O})$.

Proof. This is true because going around the diagram either way we obtain the map adjoint to the map

$$\begin{aligned} & L(f' \circ f)^* \left(R(f' \circ f)_* K \otimes^{\mathbf{L}}_{\mathcal{O}''} R(f' \circ f)_* L \right) \\ & = L(f' \circ f)^* R(f' \circ f)_* K \otimes^{\mathbf{L}}_{\mathcal{O}} L(f' \circ f)^* R(f' \circ f)_* L) \\ & \to K \otimes^{\mathbf{L}}_{\mathcal{O}} L \end{aligned}$$

in $D(\mathcal{O})$. To see this one uses that the composition of the counits like so

$$L(f' \circ f)^* R(f' \circ f)_* = Lf^* L(f')^* Rf'_* Rf_* \to Lf^* Rf_* \to \mathrm{id}$$

is the counit for $L(f' \circ f)^*$ and $R(f' \circ f)_*$. See Categories, Lemma 0DV0. □